

AURANT ACEX: A Multimodal Quantitative Trading Architecture Integrating Temporal Fusion Transformers and Mixture-of-Experts (Part 1: Abstract, Introduction, and Literature Review)

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Abstract

Financial time-series forecasting presents a formidable challenge due to extreme non-stationarity, shifting market regimes, and low signal-to-noise ratios. This paper presents AURANT ACEX, an advanced multimodal quantitative trading architecture designed for high-frequency and multi-horizon financial forecasting. Our framework seamlessly integrates a Temporal Fusion Transformer (TFT) backbone with a time-series Mixture-of-Experts (MoE) routing layer, enabling dynamic specialization across latent market regimes. By incorporating a highly optimized memory-safe data pipeline processing OHLCV data, macroeconomic indicators, and cross-sectional rankings, the model avoids lookahead bias while achieving high efficiency. The experimental results demonstrate exceptional predictive accuracy, robust uncertainty calibration, and highly stable out-of-sample portfolio performance. Through rigorous walk-forward backtesting, AURANT ACEX proves to be a highly competent, well-calibrated architecture capable of generating alpha across 1-day to 28-day prediction horizons.

1 Introduction

Quantitative finance relies heavily on the extraction of predictive signals from vast, noisy, and non-stationary datasets. Traditional econometric models, such as Autoregressive Integrated Moving Average (ARIMA) and Generalized Autoregressive Conditional Heteroskedasticity (GARCH), often fail to capture the complex, nonlinear dependencies inherent in modern market micro-structure. While deep learning models, particularly Long Short-Term Memory (LSTM) networks, have shown promise, they struggle with long-range dependencies and lack interpretability.

AURANT ACEX addresses these limitations by leveraging a state-of-the-art Temporal Fusion Transformer (TFT) architecture augmented with a time-series Mixture-of-Experts (MoE) framework. The primary innovation

lies in the system’s ability to adapt to structural breaks in market behavior. Financial markets do not operate under a single continuous distribution; they oscillate between distinct regimes, such as trending, mean-reverting, and highly volatile states.

The architecture ingests heterogeneous data streams—including OHLCV price data, derived technical indicators, and macroeconomic signals—encoding them into unified multimodal embeddings. A crucial aspect of this system is the Variable Selection Network (VSN), which dynamically weights the importance of each feature at every time step, filtering out noise and focusing on the most relevant signals. Furthermore, the model is designed with strict causal integrity, ensuring zero lookahead bias—a common pitfall in financial machine learning research.

2 Literature Review

The evolution of financial time-series forecasting has been characterized by a continuous push toward more complex, non-linear models capable of digesting diverse data modalities.

Deep Learning in Finance. Early applications of deep learning in finance focused predominantly on Multi-Layer Perceptrons (MLPs) and basic Recurrent Neural Networks (RNNs). However, these architectures suffered from the vanishing gradient problem when applied to long sequences. The introduction of LSTMs mitigated this, allowing models to retain memory over longer periods. Nonetheless, LSTMs process data sequentially, which inherently limits parallelization and struggles with extremely long-term contextual dependencies (Sezer et al., 2020).

Transformers and Attention Mechanisms. The advent of the Transformer architecture (Vaswani et al., 2017) revolutionized sequence modeling. By relying entirely on self-attention mechanisms, Transformers compute representations of a sequence simultaneously, allowing for direct modeling of dependencies regardless of distance. The

Temporal Fusion Transformer (Lim et al., 2021) specifically adapted this architecture for multi-horizon time-series forecasting by integrating static covariates, known future inputs, and past observed variables through a specialized attention mechanism.

Mixture-of-Experts (MoE). To address regime shifts in financial data, recent research has explored dynamic routing networks. MoE architectures (Shazeer et al., 2017) employ a gating mechanism to route inputs to specialized sub-networks (experts). In the context of time-series, this allows the model to develop distinct experts for different market regimes (e.g., a "momentum expert" vs. a "volatility expert"), improving generalization without exponentially increasing computational cost.

Uncertainty Estimation. Point forecasts are insufficient for robust trading systems; quantifying uncertainty is critical for risk management. Techniques such as Bayesian Model Averaging and Conformal Prediction (Romano et al., 2019) have gained traction for generating prediction intervals with finite-sample marginal coverage guarantees. AURANT ACES integrates split conformal estimators to provide statistically rigorous bounds around its forecasts.